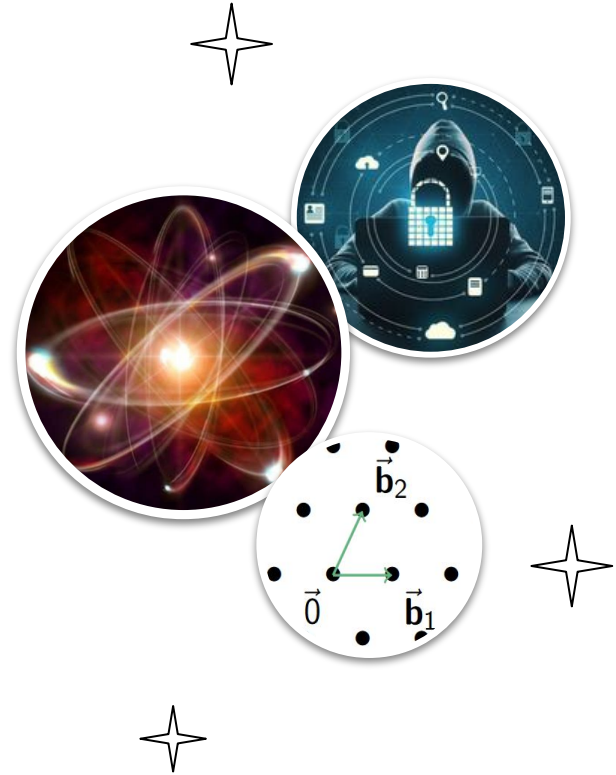


Cryptanalyse quantique des lattices

Dr. Johanna Loyer



52 av. J.C.



Chiffre de César

Chiffrer

A → B

B → C

C → D

...

Y → Z

Z → A

PREPAREZ LE PIEGE
SUR LA COLLINE
D'ALESIA

QSFQBSFA MF QJFHF
TVS MB DPMMJOF
E'BMFTJB

Déchiffrer

A → Z

B → A

C → B

...

Y → W

Z → Y



Chiffre de César

Chiffrer

A → B

B → C

C → D

...

Y → Z

Z → A

Déchiffrer

A → Z

B → A

C → B

...

Y → W

Z → Y

PVQT, QBT TJ
TFDVSJTF RVF DB !



Chiffre de César

Chiffrer

A → B

B → C

C → D

...

Y → Z

Z → A

OUPS, PAS SI
SECURISE QUE CA !

Déchiffrer

A → Z

B → A

C → B

...

Y → W

Z → Y

PVQT, QBT TJ
TFDVSJTF RVF DB !





Enigma

1939

- 83 - ADJ JNA -

LMHNX WEKLM UERDS EVHLC
 JSQQK VLDES ANEVT YEDGI
 ZQDOD RMDKG SXGSQ SHDQP
 VIEAP IENLI CLZCL LAGWC
 BJZD



Enigma

1939

- 83 - ADJ JNA -

LMHNX WEKLM UERDS EVHLC
JSQQK VLDES ANEVT YEDGI
ZQDOD RMDKG SXGSQ SHDQP
VIEAP IENLI CLZCL LAGWC
BJZD

Cryptanalyse



Enigma

1939

- 83 - ADJ JNA -

LMHNX WEKLM UERDS EVHLC
 JSQQK VLDES ANEVT YEDGI
 ZQDOD RMDKG SXGSQ SHDQP
 VIEAP IENLI CLZCL **HEIL**
HITLER

Cryptanalyse



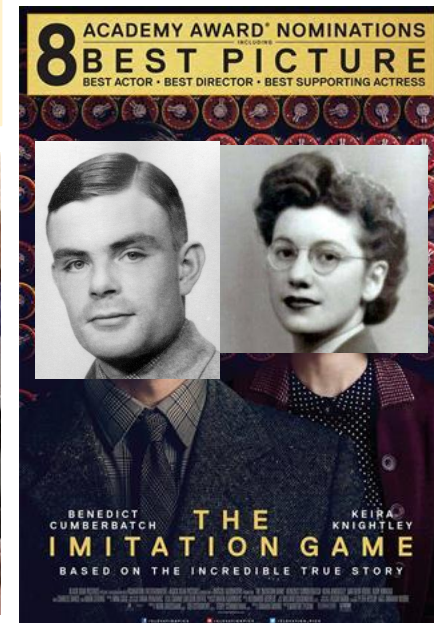
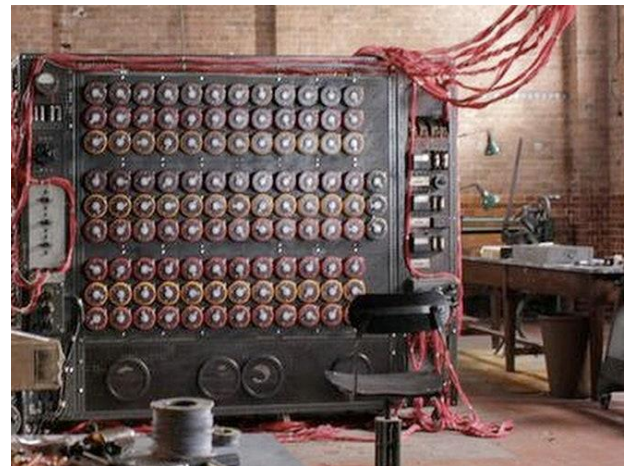
Enigma

1939

- 83 - ADJ JNA -

LMHNX WEKLM UERDS EVHLC
JSQQK VLDES ANEVT YEDGI
ZQDOD RMDKG SXGSQ SHDQP
VIEAP IENLI CLZCL **HEIL**
HITLER

Cryptanalyse





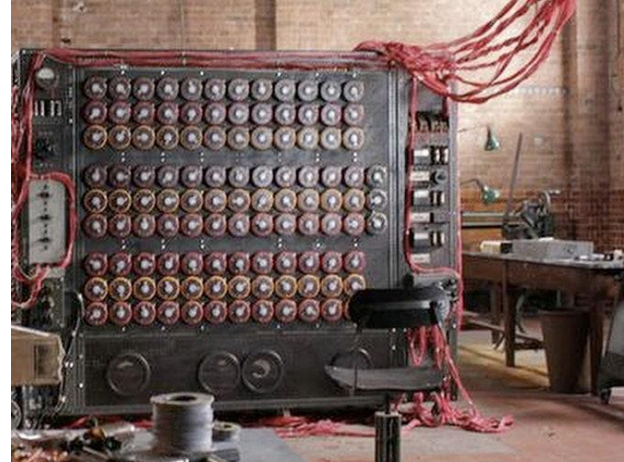
Enigma

1939

- 83 - ADJ JNA -

LMHNX WEKLM UERDS EVHLC
 JSQQK VLDES ANEVT YEDGI
 ZQDOD RMDKG SXGSQ SHDQP
 VIEAP IENLI CLZCL **HEIL**
HITLER

Cryptanalyse



Années 1970



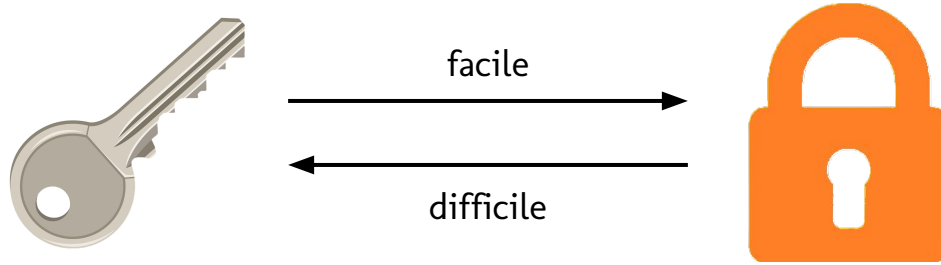
Années 1970



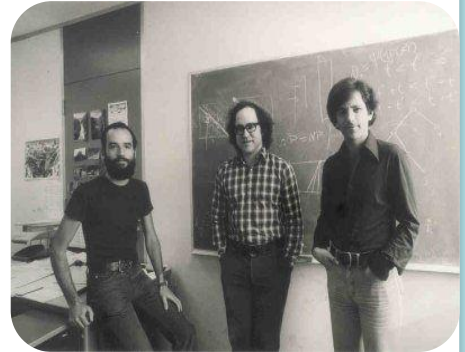
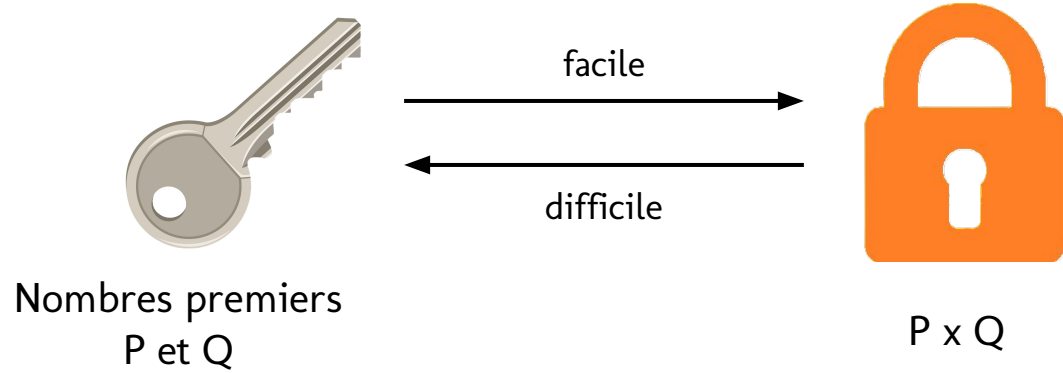
Pourquoi la sécurité informatique est-elle importante ?



Chiffrement asymétrique



Chiffrement asymétrique

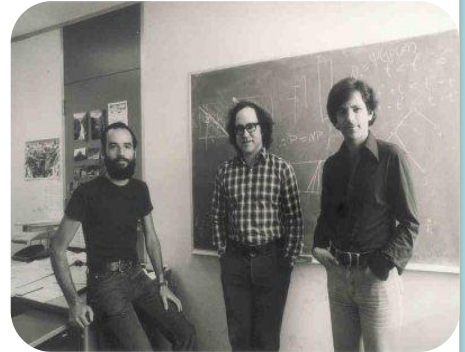
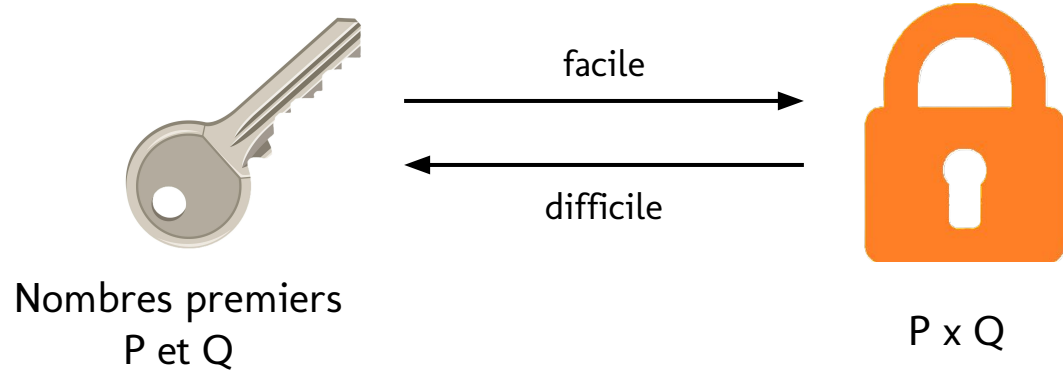


Rivest, Shamir & Adleman

$$P \times Q =$$

7487869927382228767030641005652582376664116895591925942032312045821425380552073005291076958
9310154430228567886782322253163986663671065974410591516257916784327030655524680115587855657622
69415511054634086033391917807255228011912006081719185419030150337935853314471455278465677199388
6634856241405809880117441029508975191737513326356386593323855753894172373033807560692526663128
239234718590443206457755768522058106983374764382193706546199518762495507385546703190424412194
646520350487376579137499225032384179194492563915431360553097163206866690722336841973631155010
42878337123741086644394226378033905592006616040673005608191041617047062052006640741371629
288703113244157939980609005142380531520757992544004840633308722004008246941614587623136992
44427168410010234953322369230131008875207144178060016690500990949035506940693521719866602
606841848144161385641366094783418163579176641469113458683986636271624156226808640075322348660
856868239917652135854264247202737344360664539192915544469799363161283822017014838671695003776
89640016034813328639089595188900543842473805635216548074511467632695923063570000124926420
279435942759877089278064723846752483221531798870290858806573151284453558386481970879327217338
061580998730755633028370075723805063

Chiffrement asymétrique



Rivest, Shamir & Adleman

Chiffrement asymétrique



Nombres premiers
 P et Q

facile

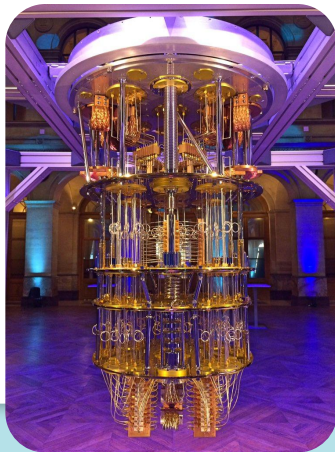


~~difficile~~

facile avec un
ordinateur quantique !



$P \times Q$



Ordi classique : 0 ou 1



Ordi quantique : 0 et 1
en même temps

Physique quantique



Particule quantique dans plusieurs
états en même temps



Chat de Schrödinger



Physique quantique



Particule quantique dans plusieurs
états en même temps



Chat de Schrödinger



« Personne ne comprend vraiment
la physique quantique. »

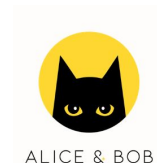
— Richard Feynman



Qui veut construire un ordinateur quantique ?



HARVARD
UNIVERSITY



Edward Snowden

Petit résumé jusqu'ici



Chiffrement

Chiffre de César



Facile à la main !

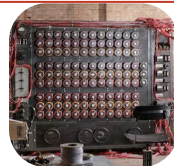
Enigma



R.S.A.
($P \times Q$)



Ordinateurs



Ordinateur quantique

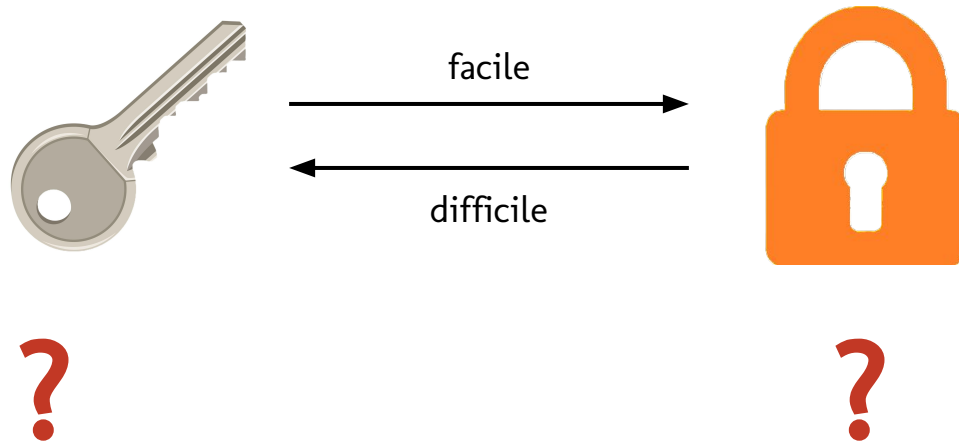


???

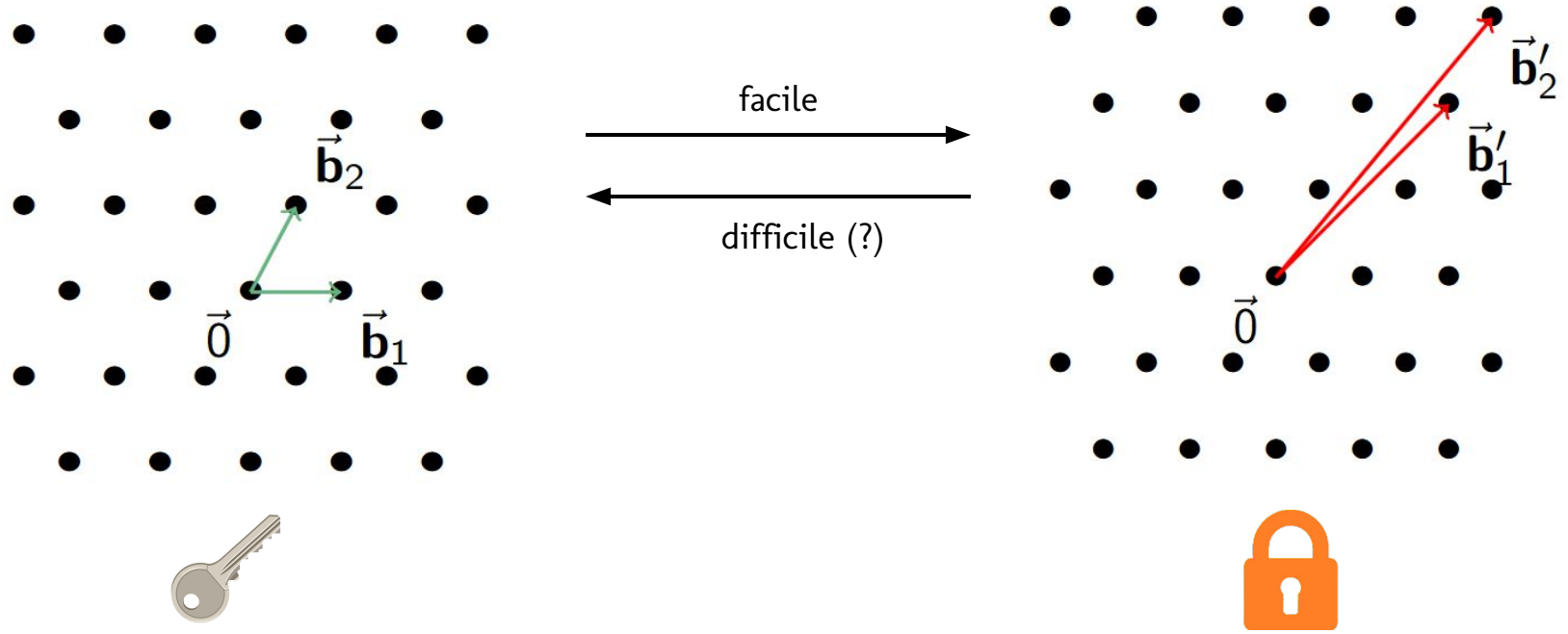


Cryptanalyse

Chiffrement “post-quantique”



Chiffrement “post-quantique”





Chiffrement

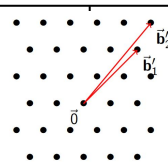
Enigma



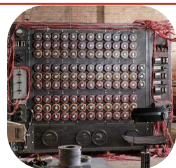
R.S.A.
($P \times Q$)



Lattices



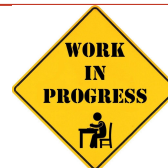
Ordinateurs



Ordinateur
quantique



???



Cryptanalyse



Les lattices sont-ils aussi
sécurisés qu'on le croit ?



Plan



1

Attaque sur les lattices

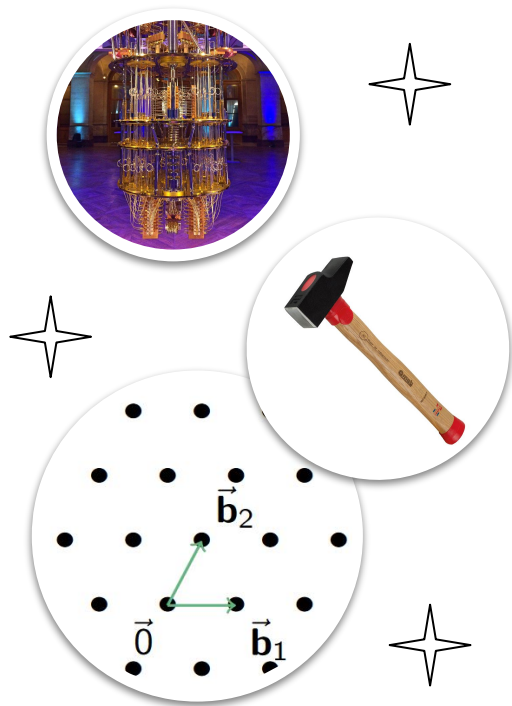
2

Amélioration de l'attaque

3

Mon attaque

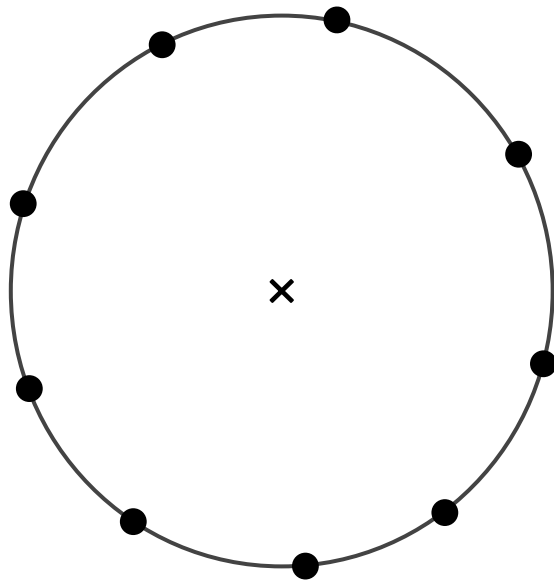




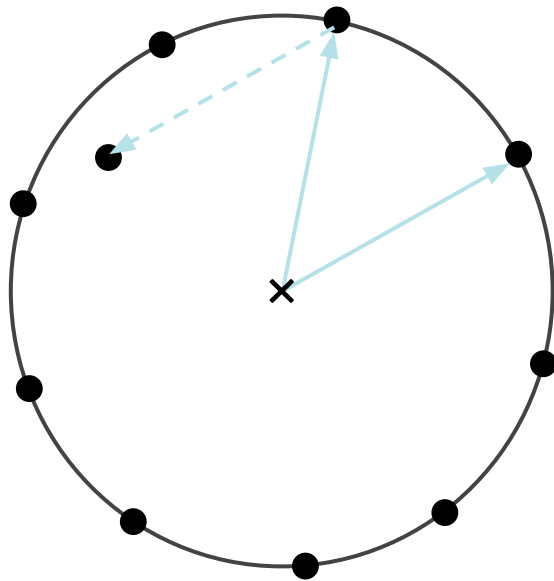
Attaque sur les lattices

En combien de temps peut-on
retrouver un petit vecteur ?

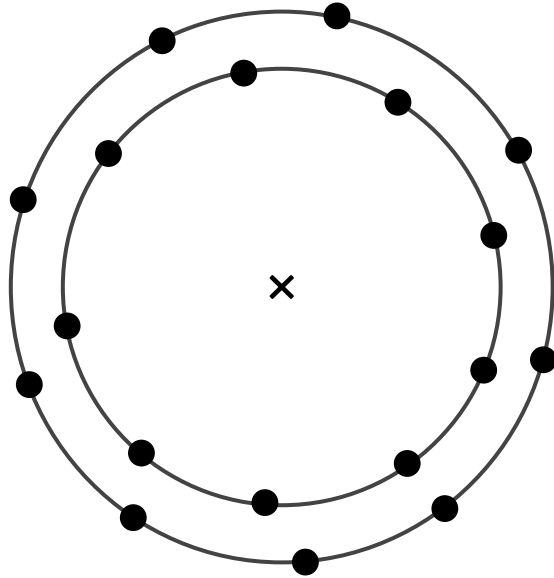
Méthode du crible



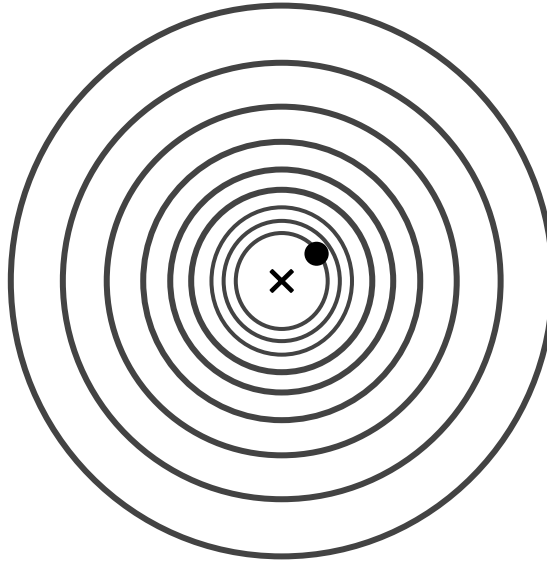
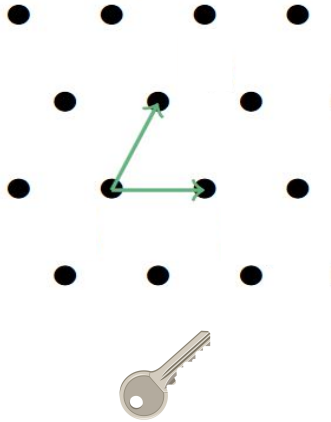
Méthode du crible



Méthode du crible



Méthode du crible



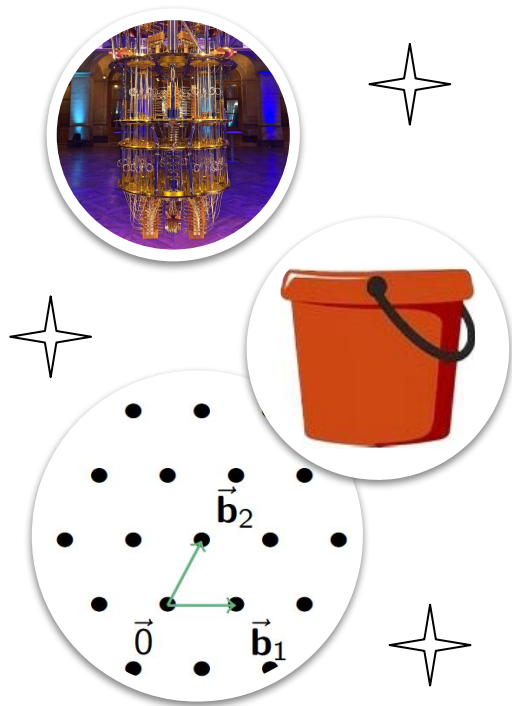
Temps de l'attaque :

$10^{0.125d}$ opérations

Meilleure attaque $> 10^{39}$:
"Sécurisé"

En dimension $d=308$:

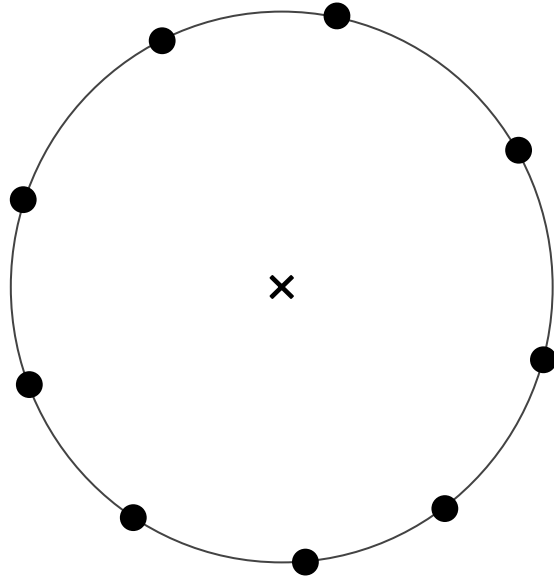
$10^{0.125d} = 10^{39}$



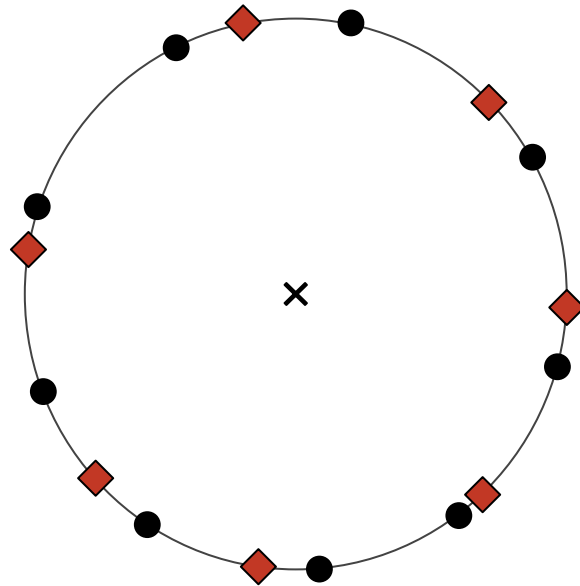
Amélioration de l'attaque

Comment trouver les points proches
plus rapidement ?

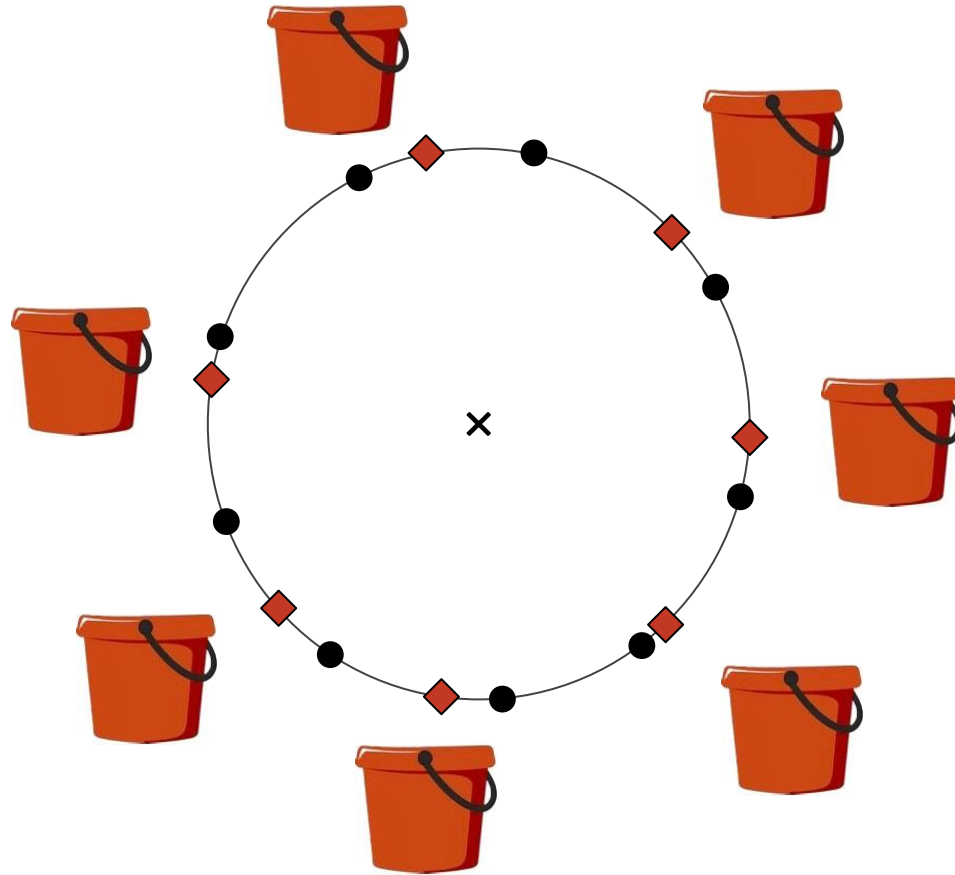
Filterage



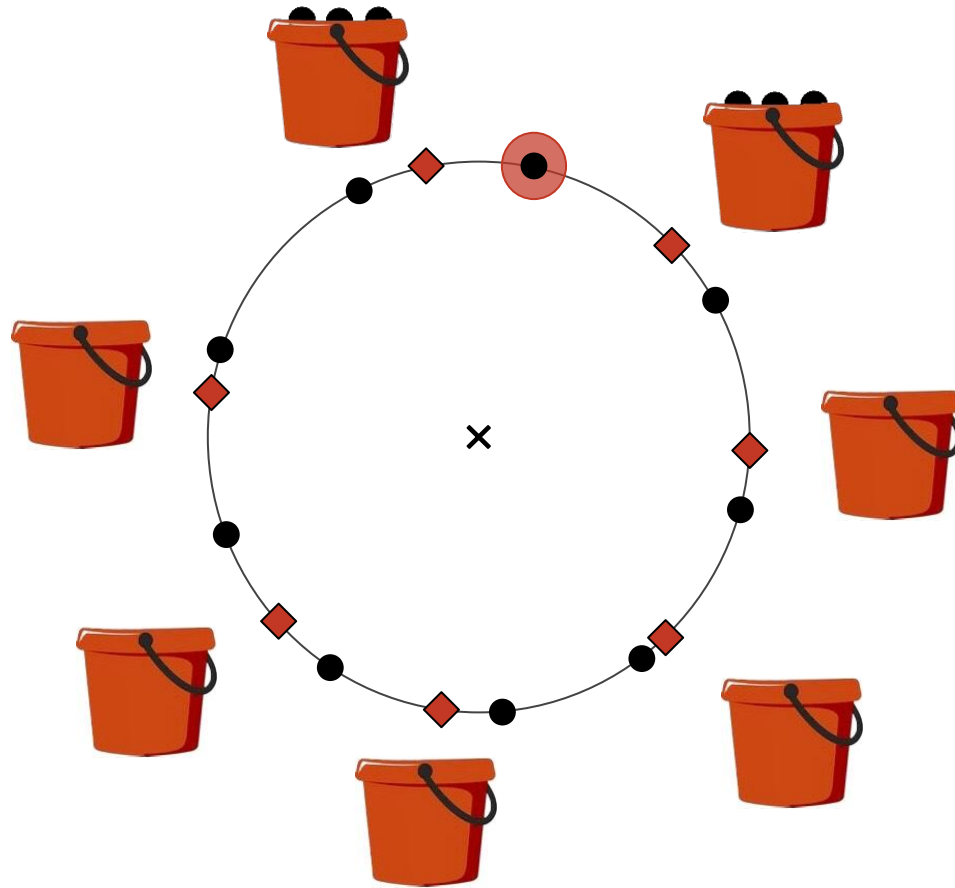
Filterage



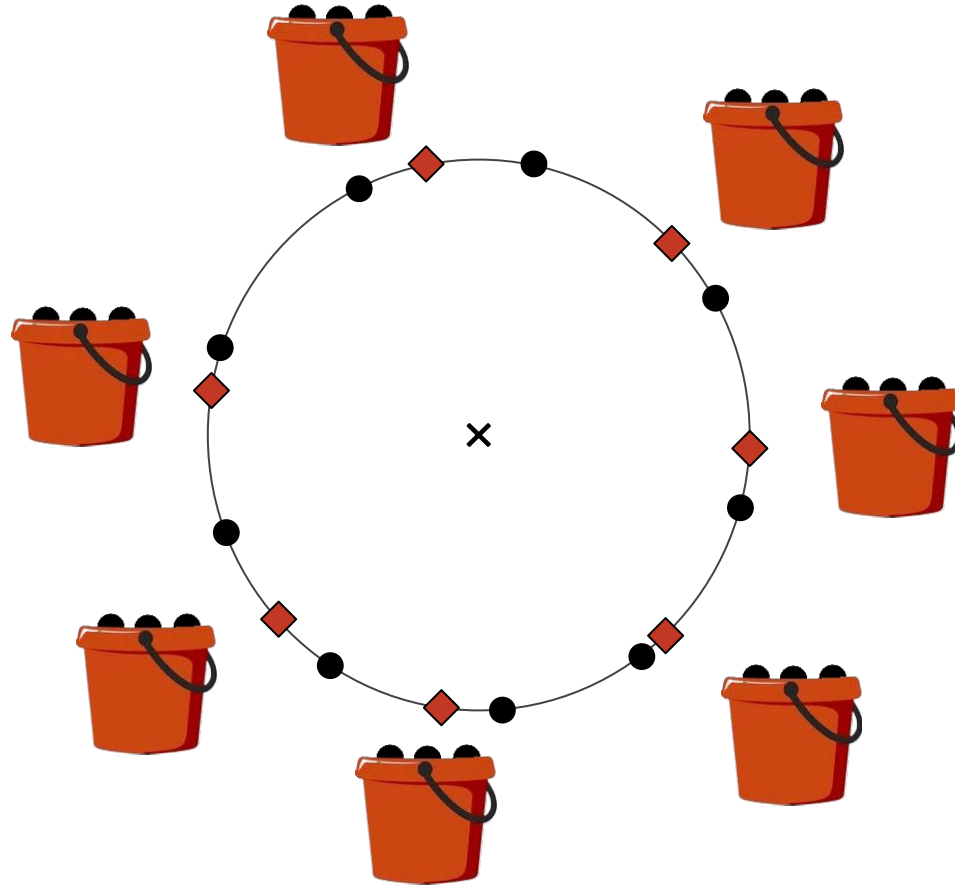
Filterage



Filterage



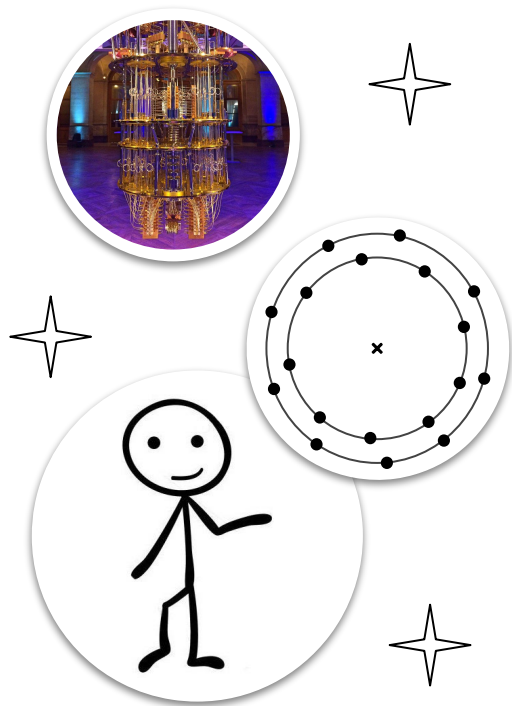
Filterage



Méthode du crible avec filtrage

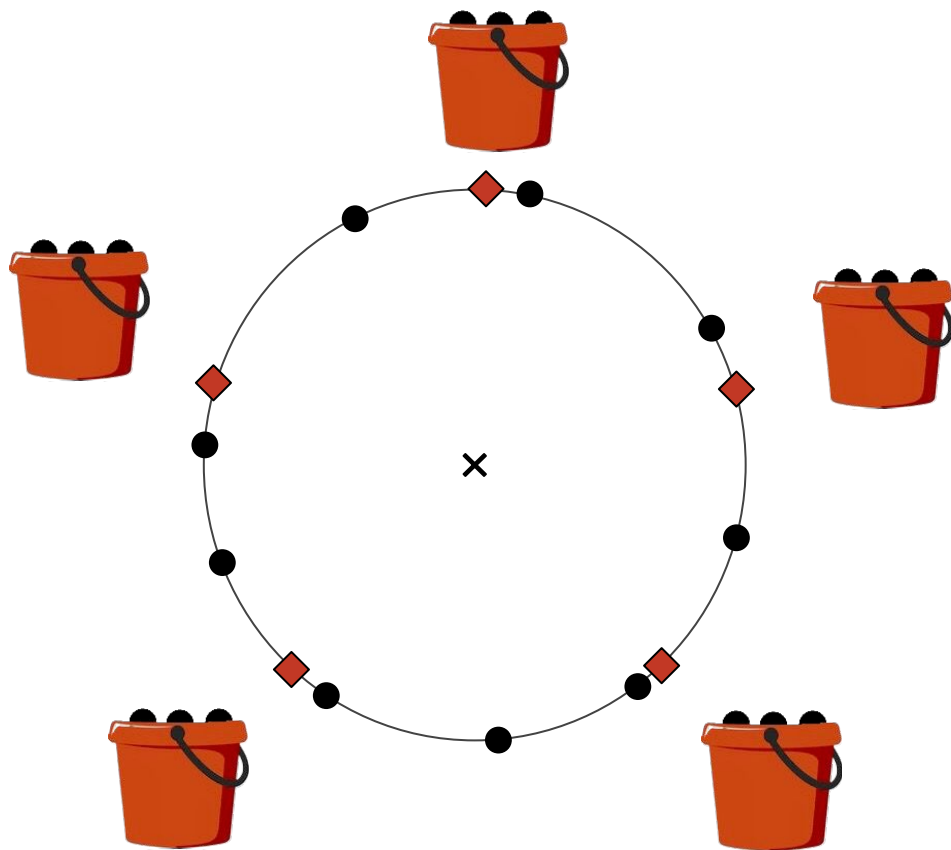
d = 500	Classique	Quantique
Sans filtrage	10^{62}	10^{47}
Avec filtrage 	10^{44}	10^{40}

Nombre d'opérations des attaques
> 10^{39} "Sécurisé"



Mon attaque

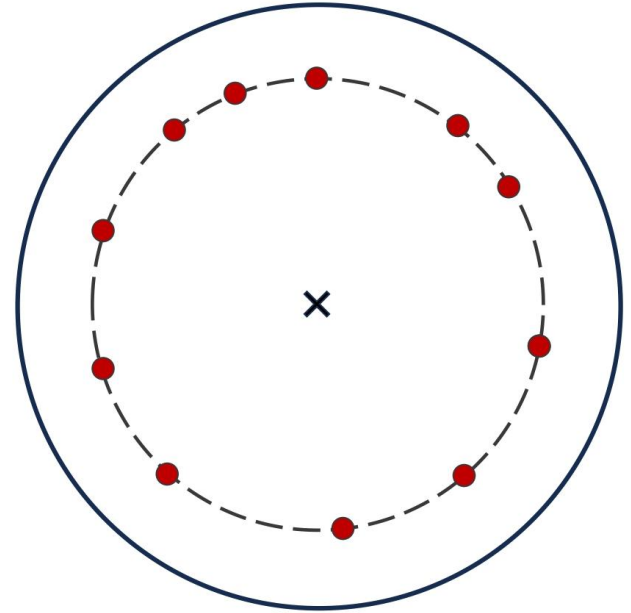
(avec Sticky en guest star) ✨



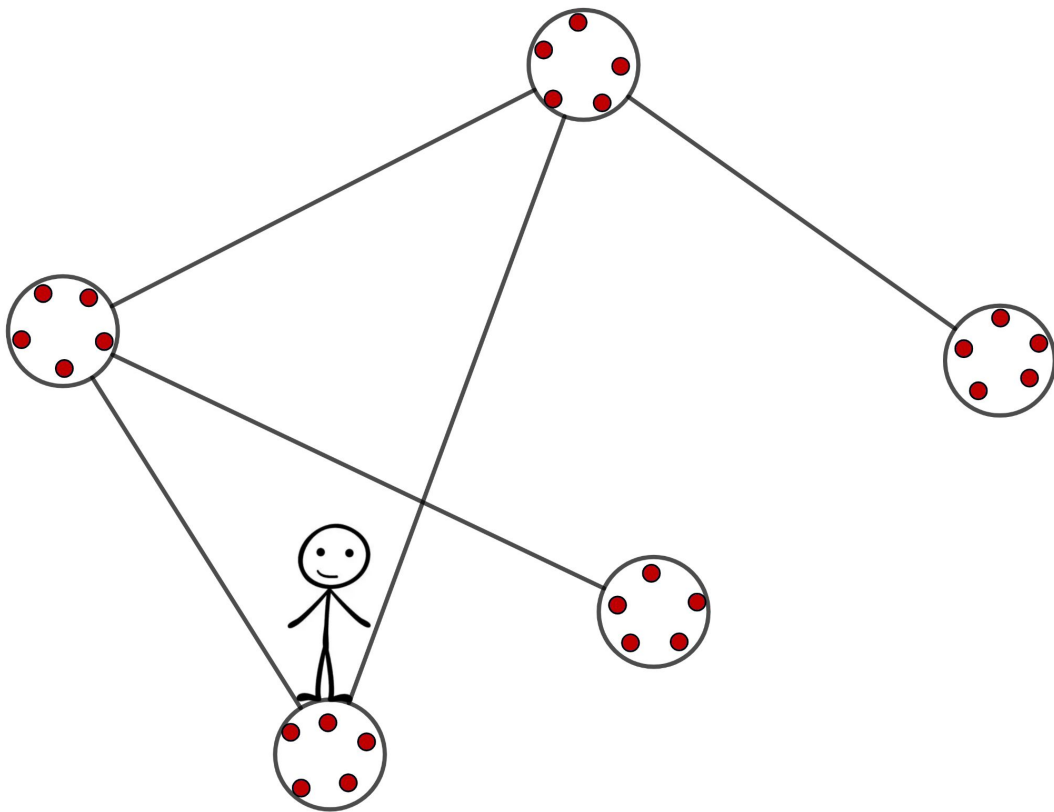
Objectif : Trouver 2 points proches dans



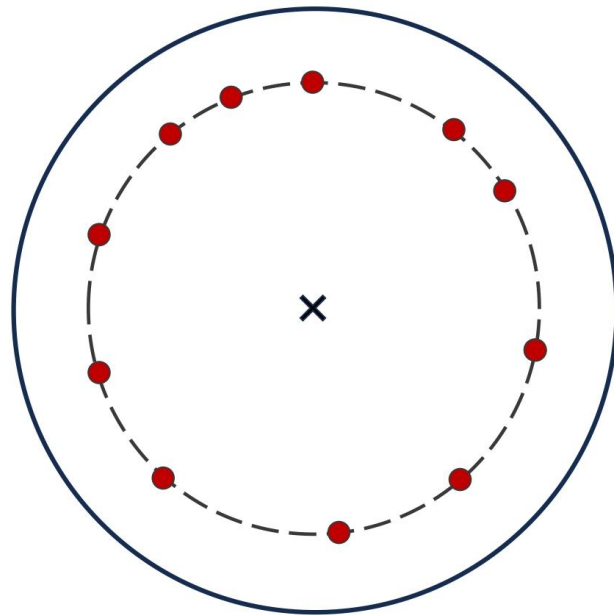
Objectif : Trouver 2 points proches dans



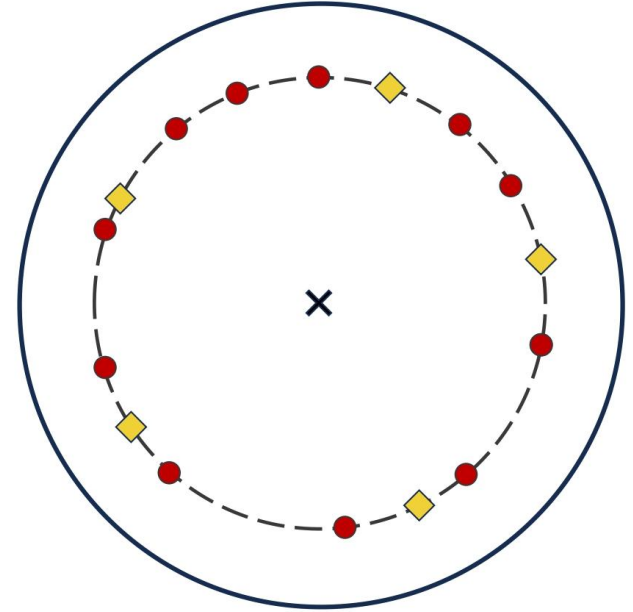
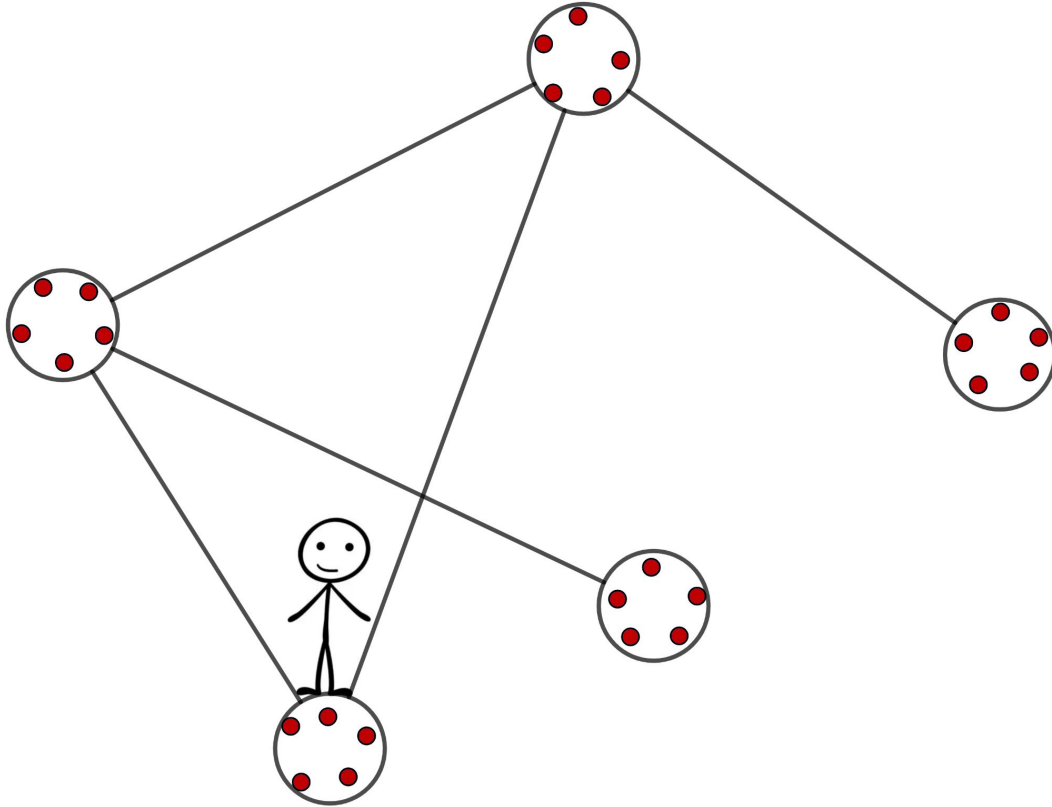
Objectif : Trouver 2 points proches dans



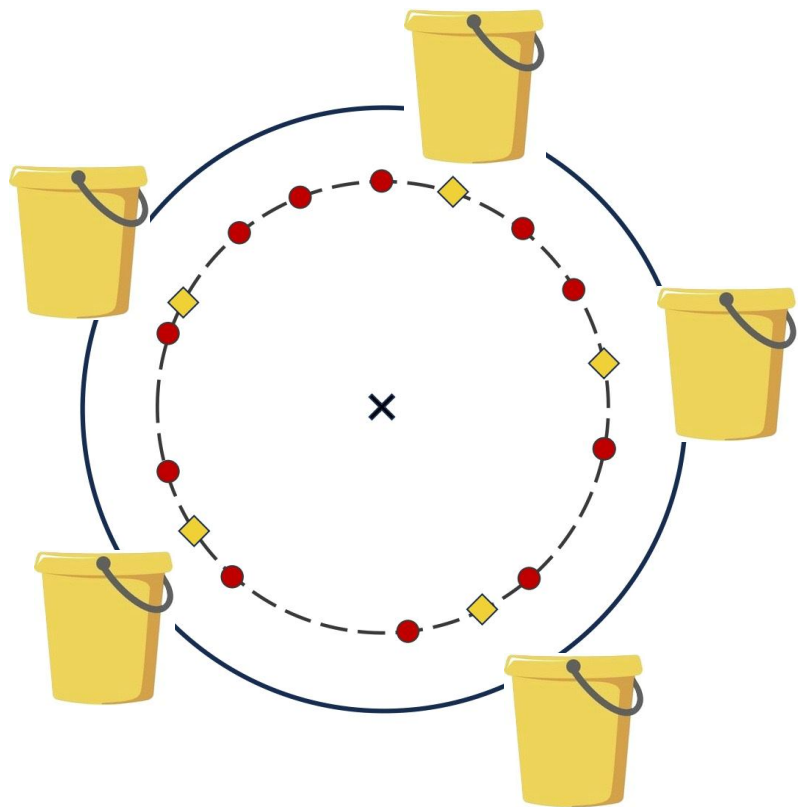
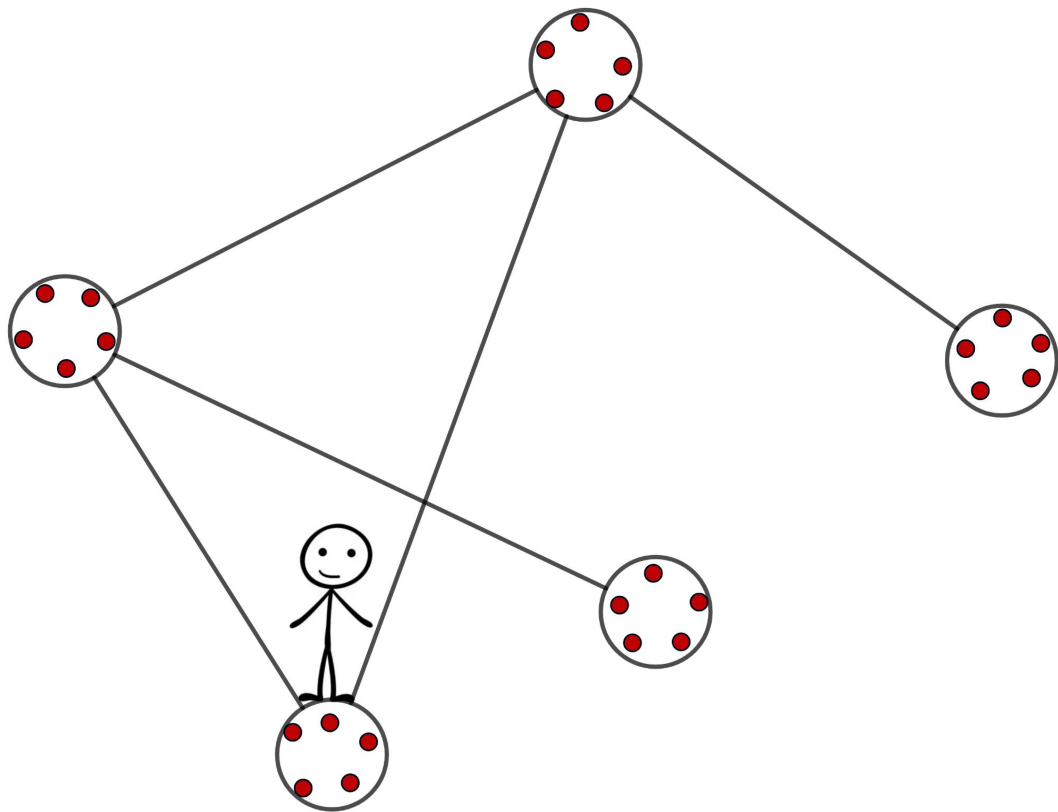
Zoom sur l'ensemble de points
où se trouve Sticky



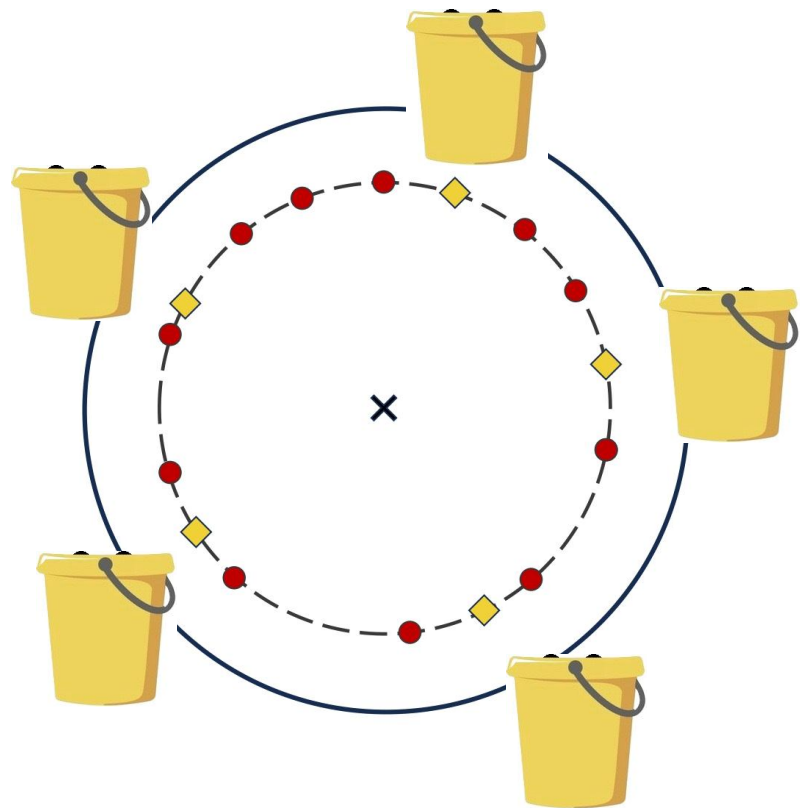
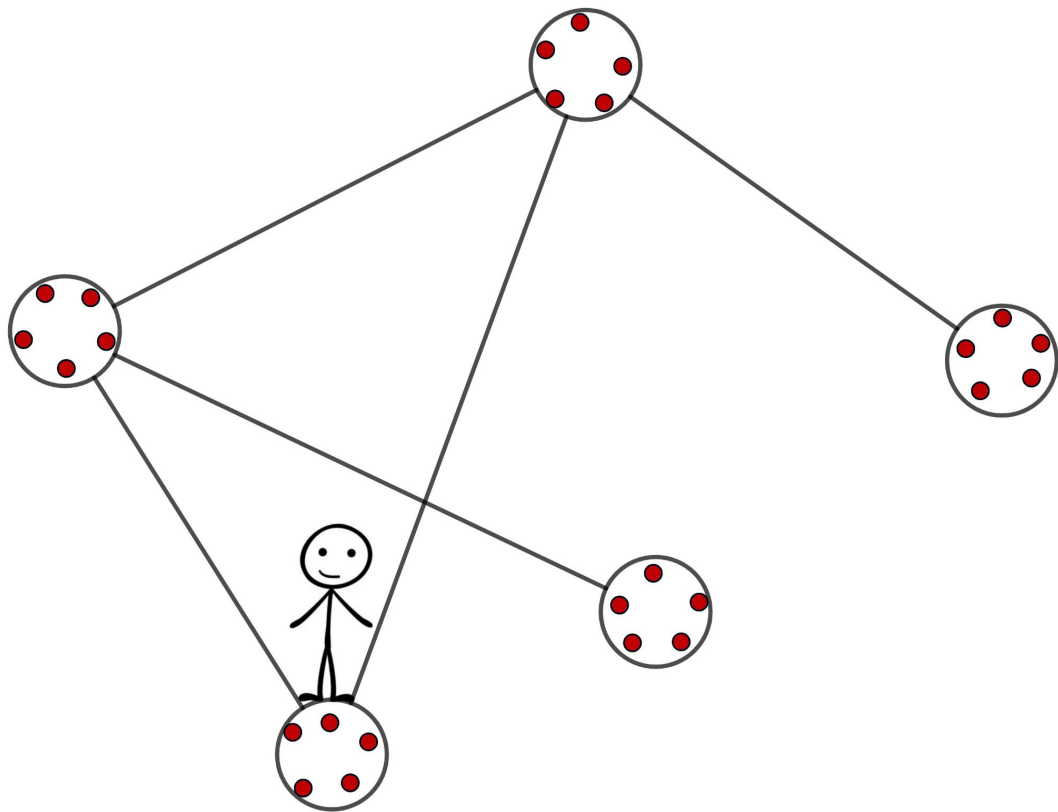
Objectif : Trouver 2 points proches dans



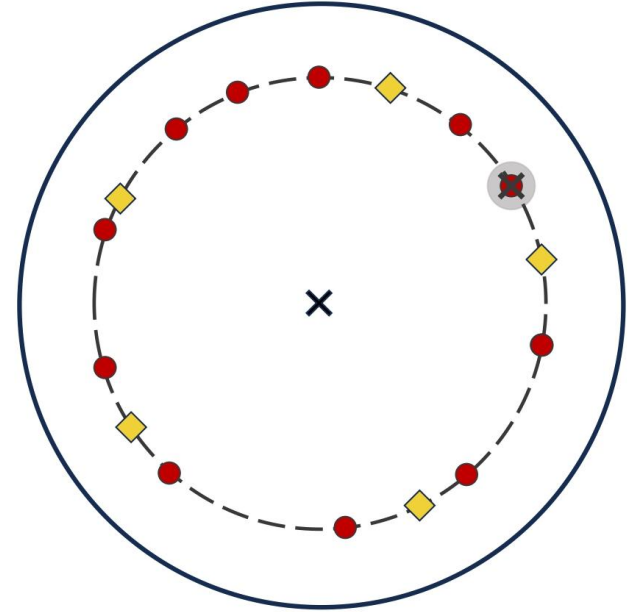
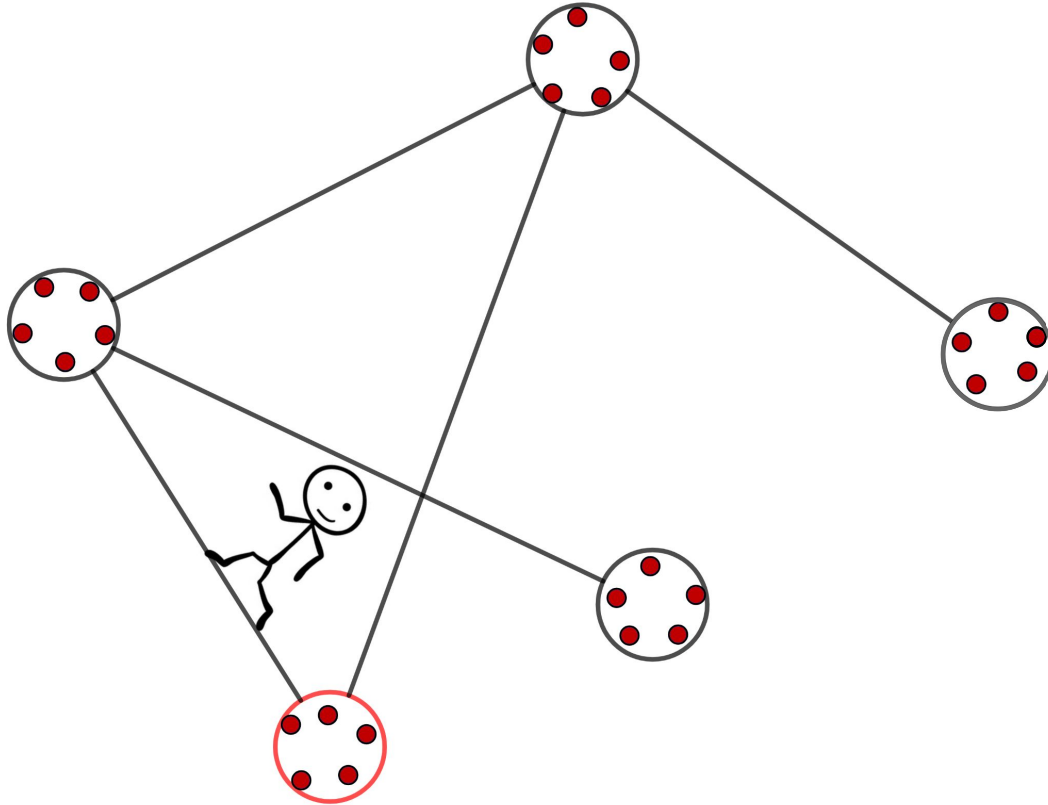
Objectif : Trouver 2 points proches dans



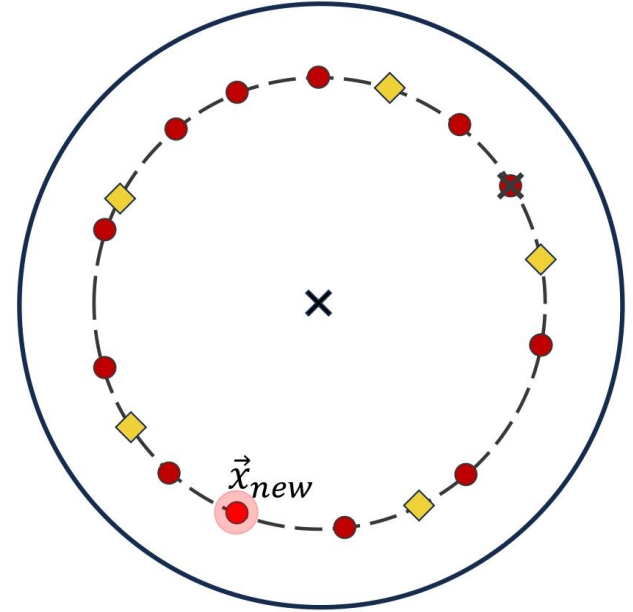
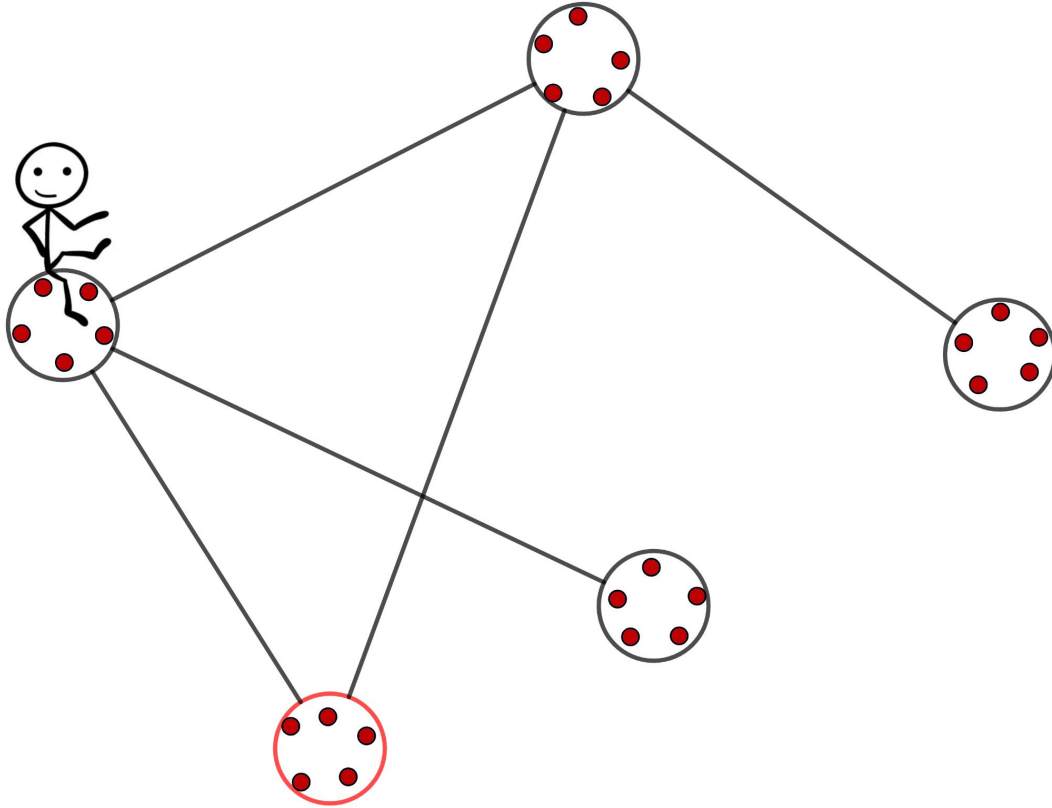
Objectif : Trouver 2 points proches dans



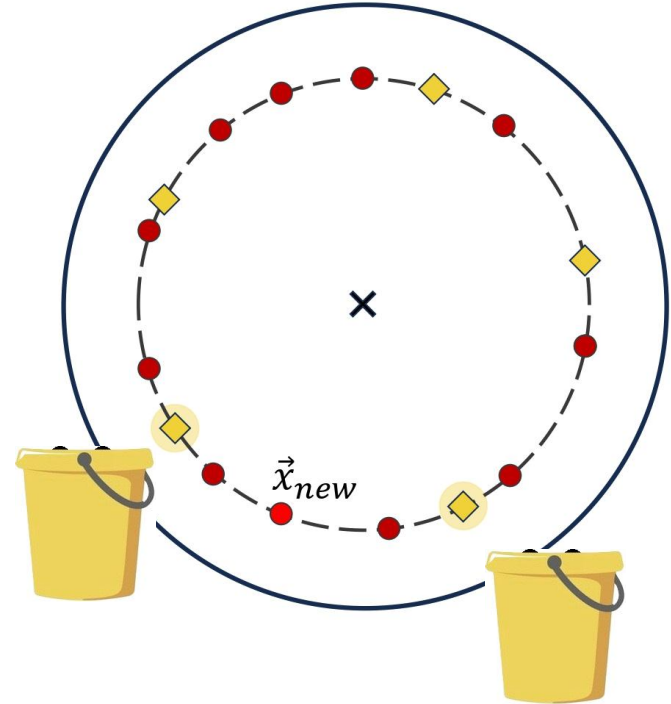
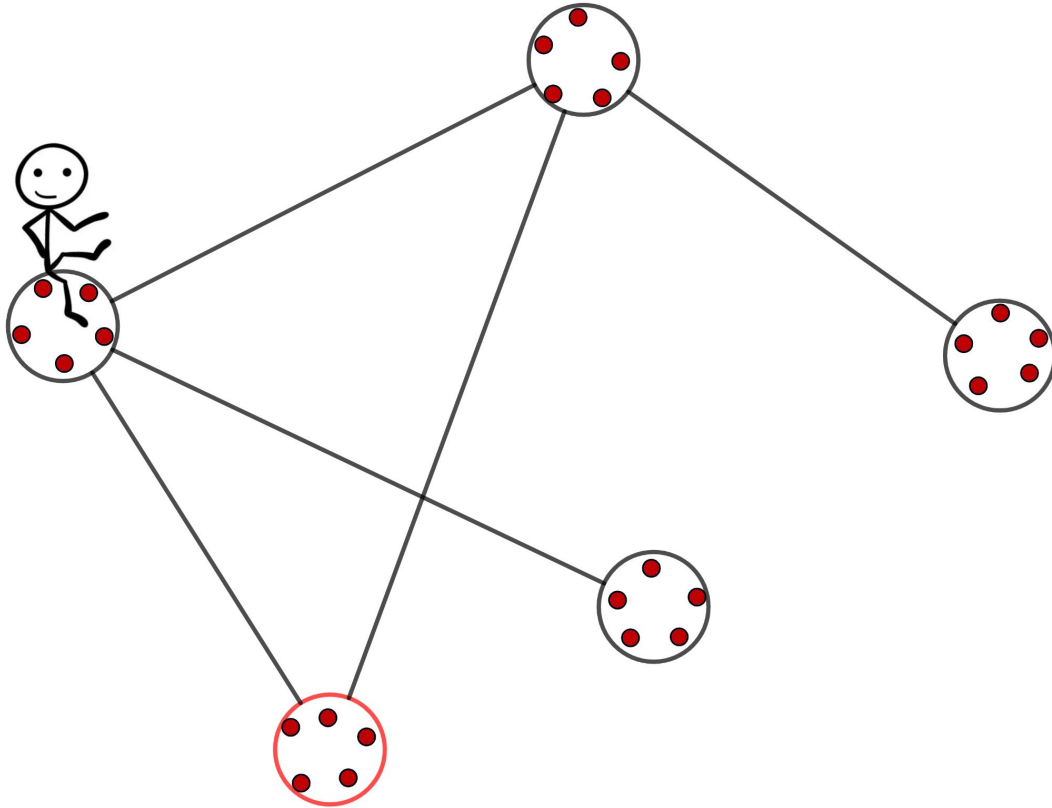
Objectif : Trouver 2 points proches dans



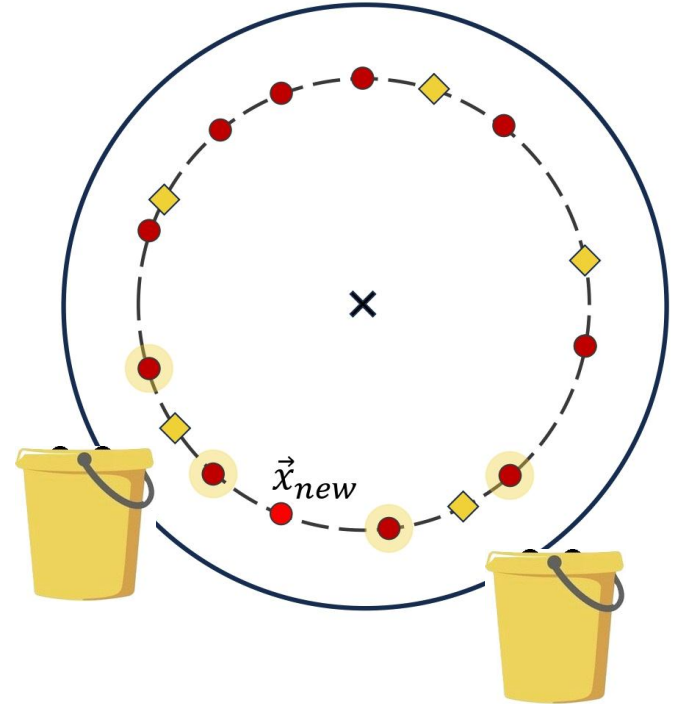
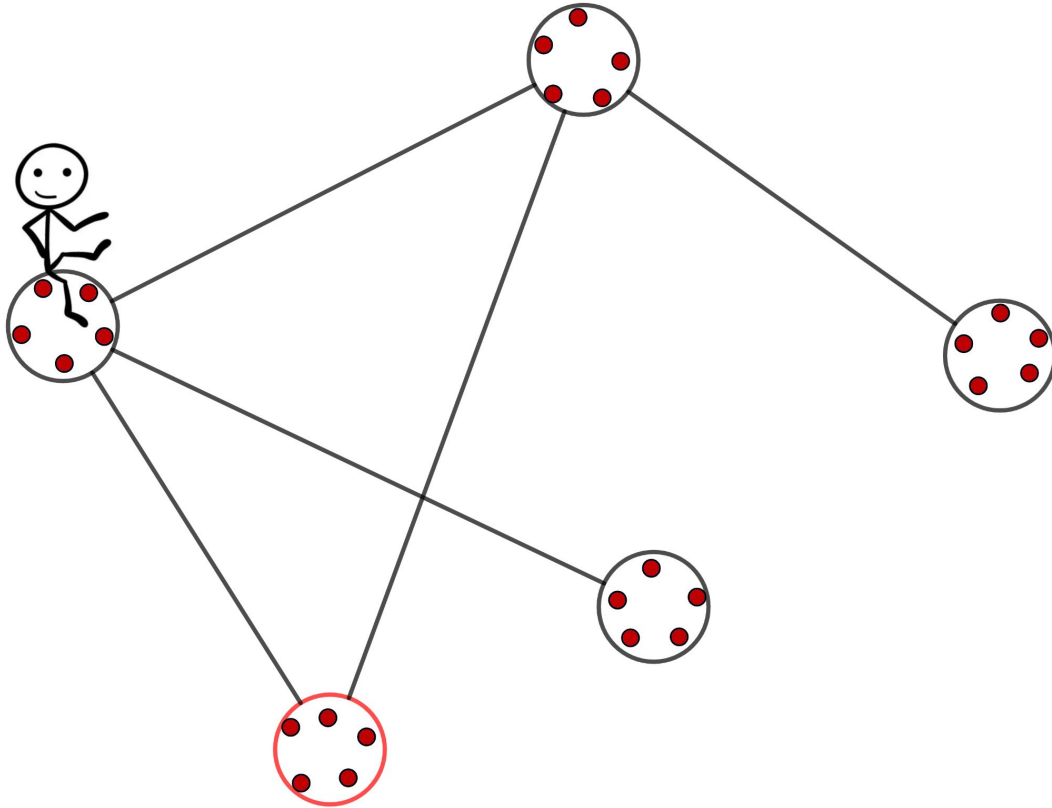
Objectif : Trouver 2 points proches dans



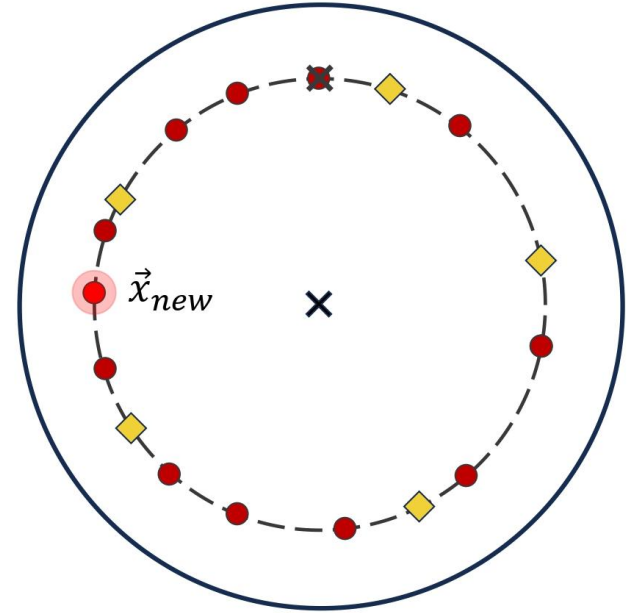
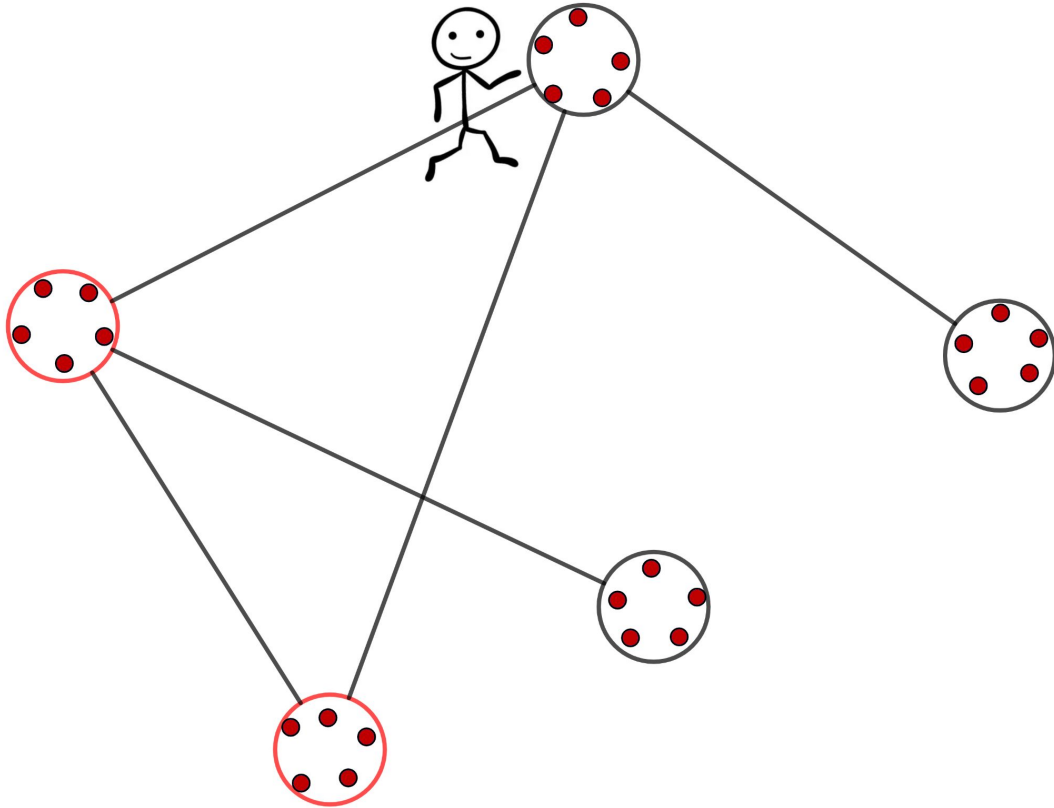
Objectif : Trouver 2 points proches dans



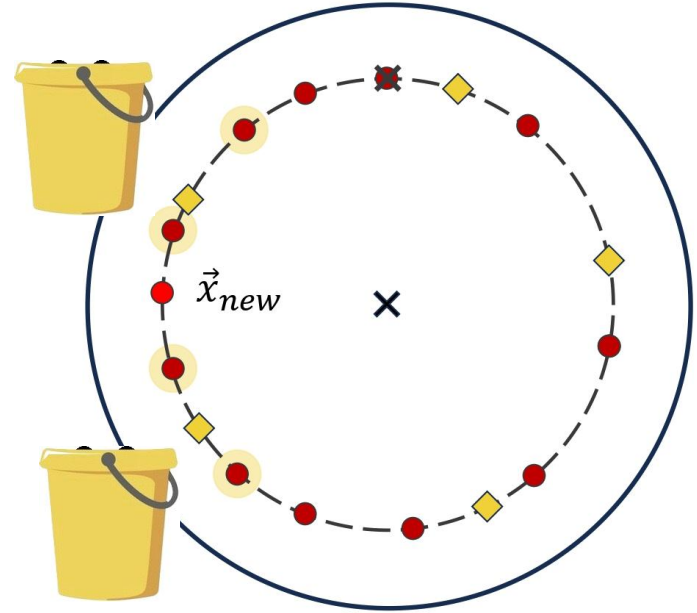
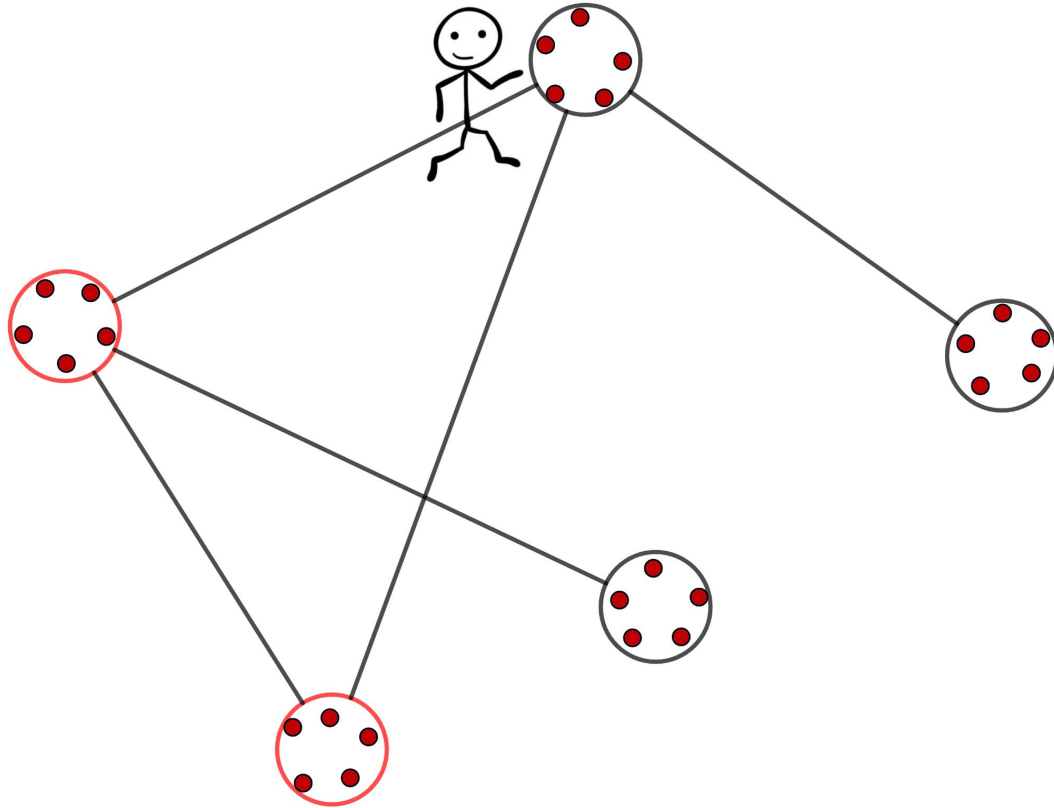
Objectif : Trouver 2 points proches dans



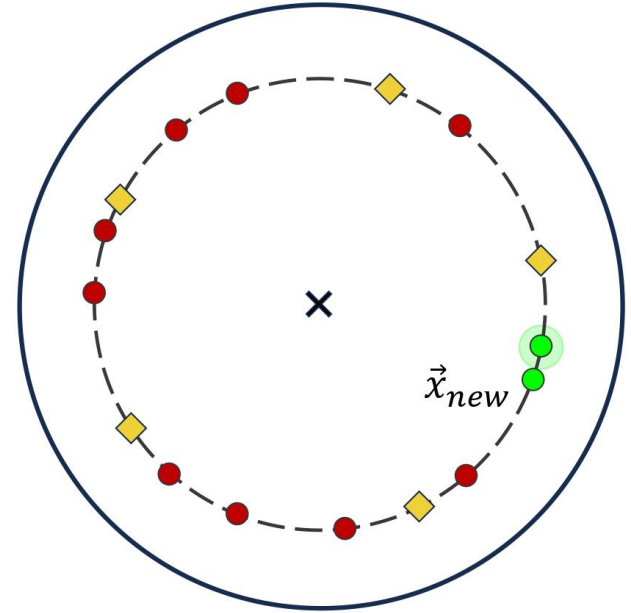
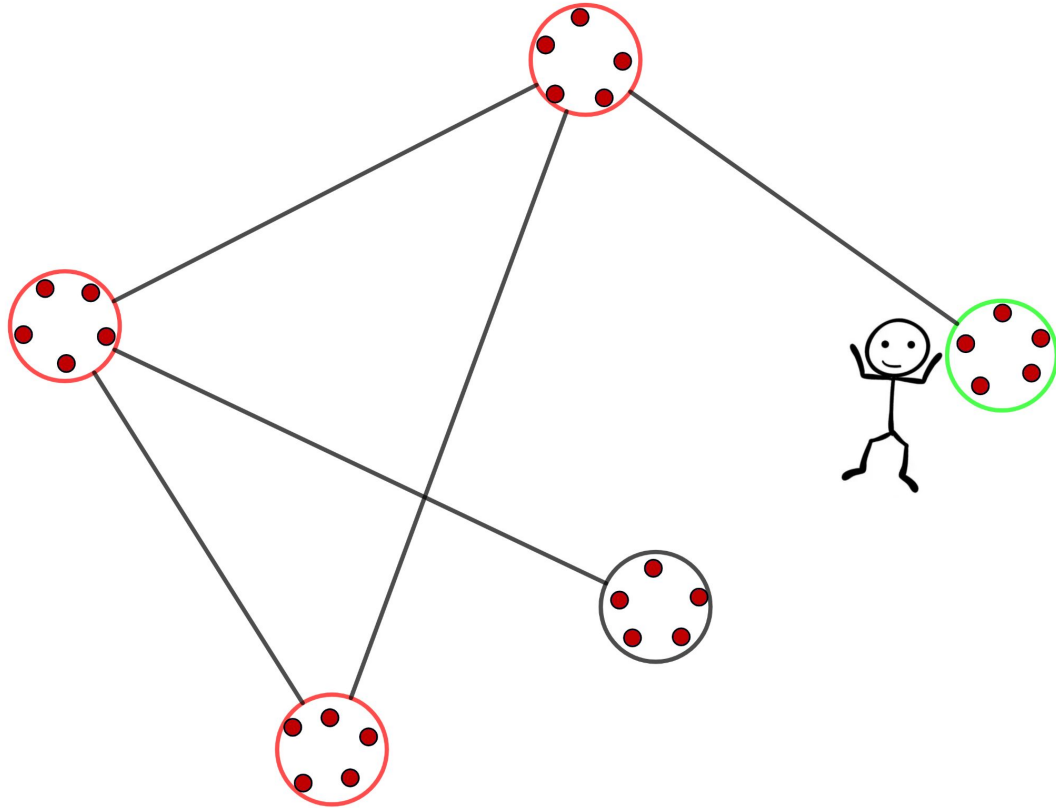
Objectif : Trouver 2 points proches dans



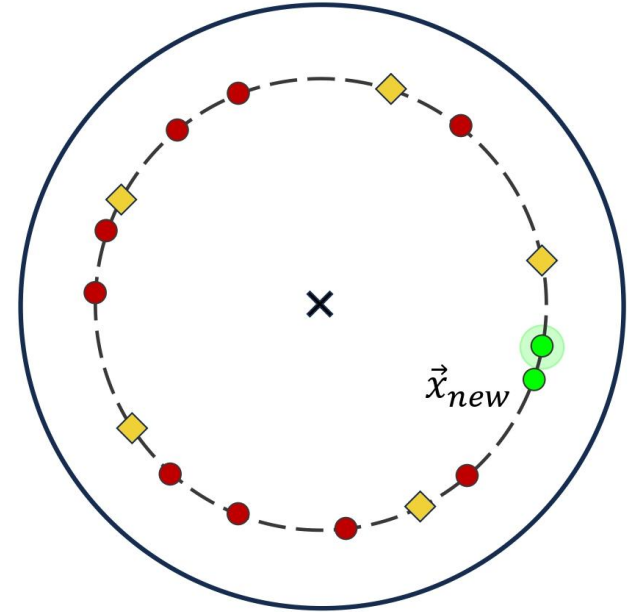
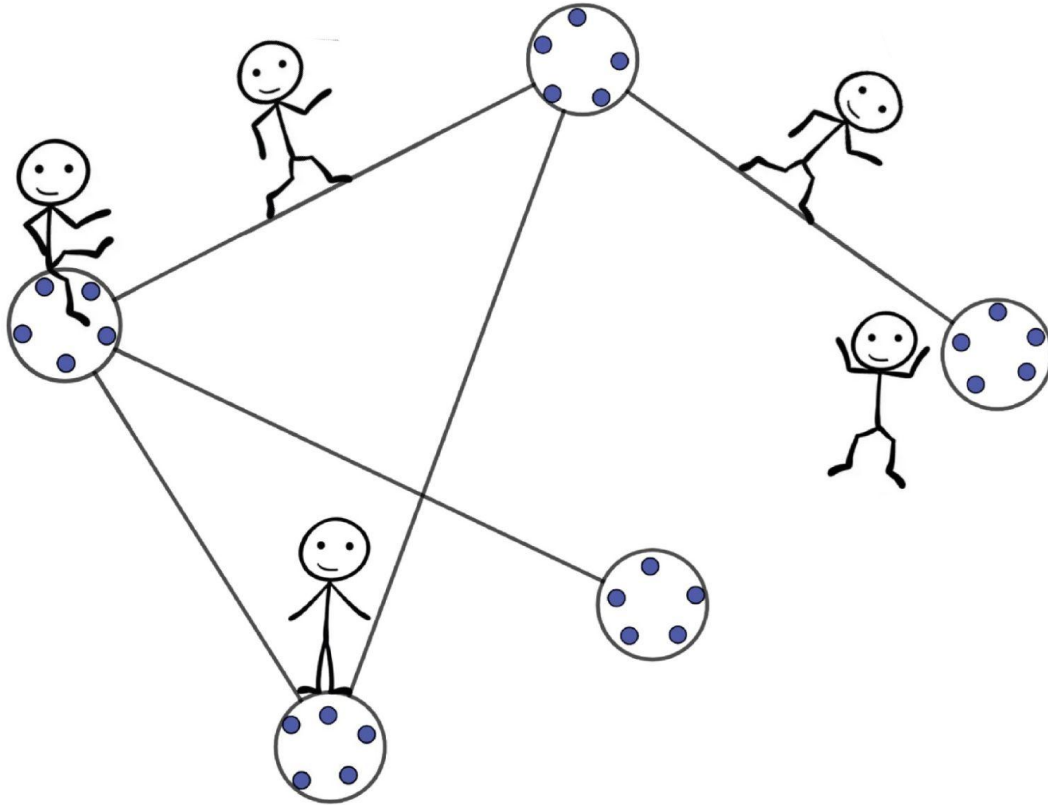
Objectif : Trouver 2 points proches dans



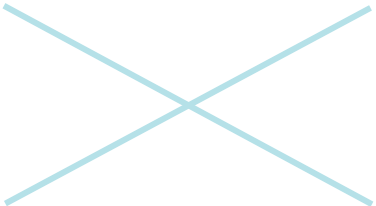
Objectif : Trouver 2 points proches dans



Objectif : Trouver 2 points proches dans

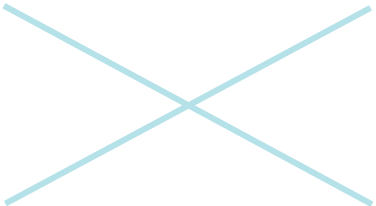


Attaques connues contre les lattices

d = 500	Classique	Quantique
Sans filtrage	10^{62}	10^{47}
Avec filtrage 	10^{44}	10^{40}
Marche quantique 		
		10^{38}



Attaques connues contre les lattices

d = 500	Classique	Quantique
Sans filtrage	10^{62}	10^{47}
Avec filtrage 	10^{44}	10^{40}
Marche quantique 		
		10^{38}
		$< 10^{39}$



Cryptanalyse quantique des lattices

Dr. Johanna Loyer

